

- Risultati di confronto tra spazi L^p
- Approssimazione di funzioni L^p con funzioni regolari
($C_0^\infty(\Omega)$, convoluzione)
- "Teorema fondamentale del calcolo in L^1 ".
(spazio delle funzioni A.C.).



• Operatori lineari

• Distribuzioni

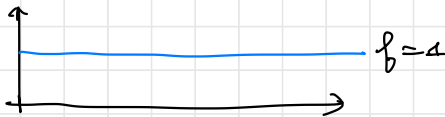
Risultati di confronto

Domanda $p < q$ $p, q \in [1, +\infty] \stackrel{?}{\Rightarrow} L^q(E) \not\subseteq L^p(E)$

Risposta: In generale NO

Controesempi $L^1(0, +\infty)$ $L^2(0, +\infty)$ $L^\infty(0, +\infty)$

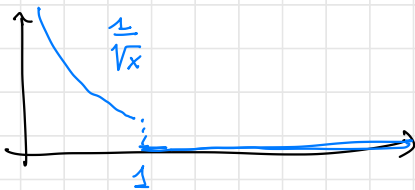
- $f \in L^\infty(\mathbb{R}_+)$ ma $f \notin L^1(\mathbb{R}_+)$, $f \notin L^2(\mathbb{R}_+)$



$$\|f\|_\infty = 1$$

$$\int_{\mathbb{R}_+} |f| = \int_{\mathbb{R}_+} |f|^2 = +\infty$$

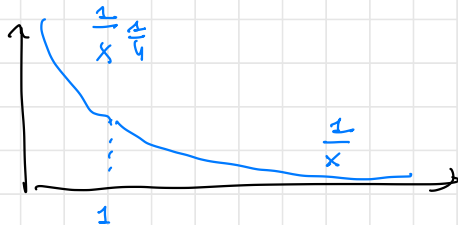
- $f \in L^1(\mathbb{R}_+)$ ma $f \notin L^2(\mathbb{R}_+)$, $f \notin L^\infty(\mathbb{R}_+)$



$$\int_0^1 \frac{1}{\sqrt{x}} < +\infty \quad \text{ma} \quad \int_0^1 \frac{1}{x} = +\infty$$

f non è m-limitata in $\mathcal{U}^+(0)$

- $f \in L^2(\mathbb{R}_+)$ ma $f \notin L^\infty(\mathbb{R}_+)$, $f \notin L^1(\mathbb{R}_+)$



$$\bullet f \notin L^\infty(\mathbb{R}_+)$$

$$\bullet f \notin L^1(\mathbb{R}_+)$$

$$\bullet f \in L^2(\mathbb{R}_+)$$

$$\int_1^{+\infty} \frac{1}{x} = +\infty$$

$$\begin{cases} \int_1^{+\infty} \frac{1}{x^2} < +\infty \\ \int_0^1 \frac{1}{\sqrt{x}} < +\infty \end{cases}$$

Disuguaglianza di Hölder

Sia $E \subseteq \mathbb{R}^n$ misurabile qualsiasi e sia $p \in [1, +\infty]$.

Sia $p' :=$ esponente coniugato di p definito da

$$p' = \frac{p}{p-1} \quad \left[\frac{1}{p} + \frac{1}{p'} = 1 \right] \quad \left(\text{con la convenzione } \frac{1}{+\infty} = 0 \right)$$

$$(+\infty)' = 1, \quad 1' = +\infty, \quad 2' = 2$$

Sia $f \in L^p(E)$, $g \in L^{p'}(E)$. Allora:

$$\|f \cdot g\|_1 \leq \|f\|_p \|g\|_{p'}$$

(in particolare, $f \cdot g \in L^1(E)$).

Esempi

$$\bullet f, g \in L^2(E) \Rightarrow \|f \cdot g\|_1 \leq \|f\|_2 \|g\|_2$$

$$\bullet f \in L^1(E), g \in L^\infty(E) \Rightarrow \|f \cdot g\|_1 \leq \|f\|_1 \|g\|_\infty$$

Infatti: $|g(x)| \leq \|g\|_\infty \quad \text{q.o. } x \in E$

$$|f(x)| |g(x)| \leq |f(x)| \|g\|_\infty \quad \text{q.o. } x \in E$$

$$\underbrace{\int_E |f(x)| |g(x)|}_{\|fg\|_1} \leq \|g\|_\infty \underbrace{\int_E |f(x)|}_{\|f\|_1}$$

Conseguenze della d.o. di Hölder

1) PROPRIETÀ DI INNESTAZIONE Sia $E \subseteq \mathbb{R}^n$ tale che $m(E) < +\infty$

Allora se $p \leq q$, si ha $\forall f \in L^q(E)$ $\Downarrow$

$$\|f\|_p \leq m(E)^{\frac{q-p}{pq}} \|f\|_q \quad L^q(E) \subseteq L^p(E)$$

(in particolare, $f \in L^p(E)$)

Commenti

• $f \in L^\infty(E) \Rightarrow f \in L^p(E) \quad \forall p \in [1, +\infty]$

$$\|f\|_p \leq m(E)^{\frac{1}{p}} \|f\|_\infty$$

$$\int_E |f|^p \leq m(E) \|f\|_\infty^p \quad (\text{elevando alla } p)$$

Dim. Sia $f \in L^q(E)$ ($q < +\infty$) $q \geq p$

$$\int_E \underbrace{|f|^p}_{\in L^{\frac{q}{p}}} \cdot \underbrace{1}_{\in (L^{\frac{q}{p}})'} \leq \int_E 1 = m(E) < +\infty$$

$\uparrow$ Hölder

$$\left(\int_E (|f|^p)^{\frac{q}{p}} \right)^{\frac{p}{q}} \cdot \left(\int_E 1 \right)^{\frac{1}{(q/p)'}} = 1 - \frac{1}{(q/p)} = 1 - \frac{p}{q} = \frac{q-p}{q}$$

$$\left(\int_E |f|^q \right)^{\frac{p}{q}} \cdot m(E)^{\frac{q-p}{q}} \quad \text{Elevo alla } \frac{1}{p} :$$

$$\|f\|_p \leq \|f\|_q \cdot m(E)^{\frac{q-p}{pq}} \quad \square$$

2) PROPRIETÀ DI INTERPOLAZIONE

Se $E \subseteq \mathbb{R}^n$ misurabile qualsiasi e siano $p < q$.

Sia $f \in L^p(E) \cap L^q(E)$, allora

$f \in L^r(E)$ per ogni $p \leq r \leq q$, e vale

$$\|f\|_r \leq \|f\|_p^\alpha \cdot \|f\|_q^{1-\alpha}$$

(dove $\alpha \in (0, 1)$ tale che $\frac{1}{r} = \frac{\alpha}{p} + \frac{(1-\alpha)}{q}$).

Teorema di approssimazione in L^p con funzioni "regolari"

Sia Ω aperto $\subseteq \mathbb{R}^n$ e sia $p \in [1, +\infty)$

Allora $\underbrace{C_0^\infty(\Omega)}_{\text{" "}}$ è denso in $L^p(\Omega)$. $\overline{C_0^\infty(\Omega)} = L^p(\Omega)$
 $\overline{\Omega} = \mathbb{R}$

$\{f: \Omega \rightarrow \mathbb{R} \text{ di classe } C^\infty(\Omega) \text{ e aventi supporto compatto in } \Omega\}$

ovvero:

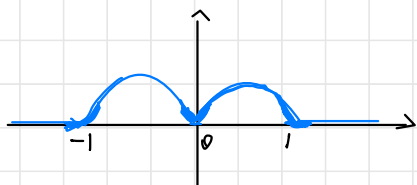
• $\forall f \in L^p(\Omega), \forall \varepsilon > 0, \exists \psi \in C_0^\infty(\Omega)$ tale che
 $\|f - \psi\|_p < \varepsilon \quad \left(\left(\int_\Omega |f - \psi|^p \right)^{1/p} < \varepsilon \right).$

• $\forall f \in L^p(\Omega) \exists \{\psi_n\} \subseteq C_0^\infty(\Omega)$ tale che
 $\psi_n \xrightarrow{L^p} f$ cioè $\lim_{n \rightarrow +\infty} \|\psi_n - f\|_p = 0$

Def. Sia $\psi \in C^\infty(\Omega)$,

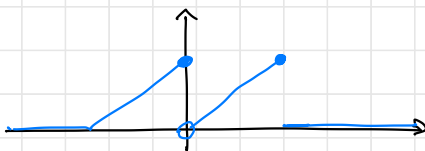
$$\text{supp } \psi := \overline{\{x \in \Omega: \psi(x) \neq 0\}}$$

Esempi $\Omega = \mathbb{R}$



$$\{\psi_1 \neq 0\} = (-1, 0) \cup (0, 1)$$

$$\text{supp } \psi_1 = [-1, 1]$$



$$\{\psi_2 \neq 0\} = (-1, 1]$$

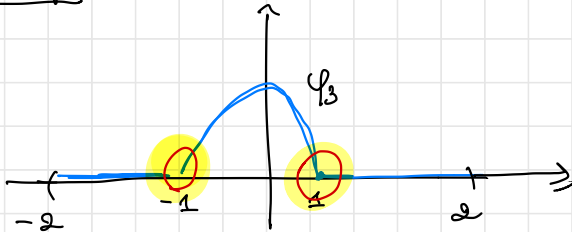
$$\text{supp } \psi_2 = [-1, 1]$$

$$\psi_1 \stackrel{?}{\in} C_0^\infty(-1, 1) \quad \text{NO!}$$

$$\psi_1 \in C_0^\infty(-2, 2) \quad \text{SÌ!}$$

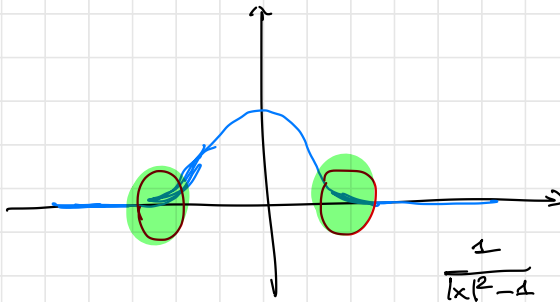
$$\psi_2 \notin C_0^\infty(-2, 2) \quad \text{NO!}$$

Esempio



$$\psi_3 \stackrel{?}{\in} C_0^\infty(-2, 2) \quad \text{NO}$$

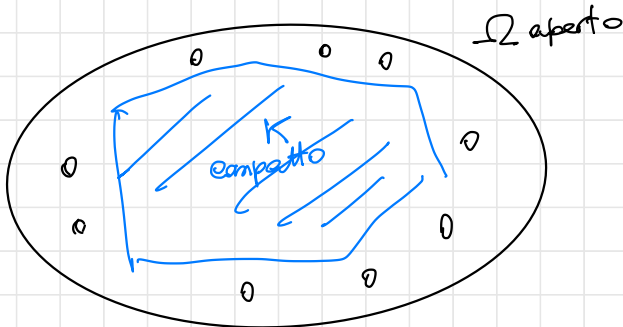
$$\text{Supp } \psi_3 = [-1, 1]$$



$$\psi_4(x) = \begin{cases} e^{\frac{1}{1-x^2-1}} & x \in (-1, 1) \\ 0 & x \in (-2, 2) \setminus (-1, 1) \end{cases}$$

$$x \in (-1, 1)$$

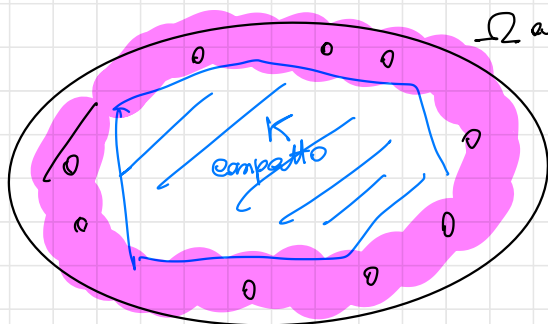
$$x \in (-2, 2) \setminus (-1, 1)$$



Ans.

Il teorema è falso per $p = +\infty$.

Ω aperto



$$f \equiv 1$$

$$f \in L^\infty(\Omega)$$

$$\|f - \varphi\|_\infty \geq 1$$

Prodotto di convoluzione

Ques. $f, g \in L^1(E)$ ~~$\Rightarrow$~~ $f \cdot g \in L^1(E)$

$$E = (0, 1) \quad f = g = \frac{1}{\sqrt{x}} \quad \text{ma} \quad f \cdot g = \frac{1}{x} \notin L^1(E)$$

Proposizione 1 Siano $f, g \in L^1(\mathbb{R})$. Poniamo

$$f * g(x) := \int_{\mathbb{R}} f(x-y)g(y) dy \quad x \in \mathbb{R}$$

Allora

(i) $f * g(x)$ esiste per q.o. $x \in \mathbb{R}$ $\leftarrow$

(ii) $f * g \in L^1(\mathbb{R})$ $\leftarrow$

(iii) $\|f * g\|_1 \leq \|f\|_1 \|g\|_1$ $\leftarrow$

Dim.

(i) $H(x, y) := f(x-y)g(y) \quad (x, y) \in \mathbb{R}^2$

A priori, non sappiamo se $H \in L^1(\mathbb{R}^2)$

$\Rightarrow$ non possiamo applicare Fubini

$\Rightarrow$ considero $|H(x-y)| = |f(x-y)||g(y)|$

Possiamo applicare Tonelli?

$$\begin{aligned} \bullet \int_{\mathbb{R}_x} |H(x, y)| dx &= \int_{\mathbb{R}_x} |f(x-y)| |g(y)| dx \\ &= |g(y)| \cdot \underbrace{\int_{\mathbb{R}_x} |f(x-y)| dx}_{\|f\|_1} < +\infty \end{aligned}$$

$$\bullet \int_{\mathbb{R}_y} \left[\int_{\mathbb{R}_x} |H(x,y)| dx \right] dy = \int_{\mathbb{R}_y} |g(y)| \|f\|_1 dy$$

$$= \|f\|_1 \|g\|_1 < +\infty$$

$$\Rightarrow H \in L^1(\mathbb{R}_x \times \mathbb{R}_y)$$

$\Rightarrow$ posso applicare Fubini

$\Rightarrow$ vale (i) perché per q.o. $x \in \mathbb{R}$

$$\int_{\mathbb{R}} H(x,y) dy < +\infty$$

(ii) segue da (iii)

$$(iii) \boxed{\|f * g\|_1} = \int_{\mathbb{R}_x} |f * g(x)| dx =$$

$$= \int_{\mathbb{R}_x} \left| \int_{\mathbb{R}_y} f(x-y) g(y) dy \right| dx \stackrel{\text{magg. modulo}}{\leq} \int_{\mathbb{R}_x} \int_{\mathbb{R}_y} \underbrace{|f(x-y) g(y)|}_{|H(x,y)|} dy dx$$

$$\stackrel{\text{Fubini}}{=} \int_{\mathbb{R}_y} \int_{\mathbb{R}_x} |f(x-y)| |g(y)| dx dy = \int_{\mathbb{R}_y} |g(y)| \|f\|_1 dy$$

$$= \boxed{\|f\|_1 \|g\|_1}$$

□